Jovan Cicvaric

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Skills

ML & Data
Python, PyTorch, NumPy, pandas, MLflow, Scikit-learn, Optuna, Keras, Wandb, Triton Inference, TensorRT, ONNX

Backend & MLOps
Kubernetes, PostgreSQL, RabbitMQ, Flask, SQLite, Helm Charts, Git, Docker, MongoDB, Celery

Frontend
React, HTML, CSS

Languages: English (Fluent), German (Intermediate), Serbian (Native), Russian (Native)

Work Experience

Optocycle GmbH, Machine Learning Engineer — Germany, 02/2024 – Present

- Developed and deployed a computer vision model for construction waste classification, achieving over 90% accuracy and improving sorting efficiency.
- Engineered an object size estimation model using monocular and stereo camera inputs, successfully detecting the largest object in over 70% of cases.
- Designed and implemented the full data labelling pipeline for 80K$+$ RGB images, and communicated with experts and customers to define annotation requirements.
- Led the development of a novel plastics classification model by combining multispectral vision with an LLM-based approach, utilizing a newly created dataset from custom multispectral sensors.
- Conducted comparative studies on classification performance that demonstrated the superior performance of RGB data over multispectral data for this specific task.

Autonomous Vision Group, Research Assistant -> Research Engineer (part-time since 02/2024) — Germany, 06/2021 – Present

- Led full-stack development of the scientific paper recommendation system, scaling the platform to support over 20K new users.
- Optimized SQL queries and the embedding storage system, achieving a $5\times$ improvement in latency and memory usage.
- Engineered the personalized recommendation system, deploying models trained on user votes and selected authors against a database of 3 million scientific articles.

Education

MS	**University of Tuebingen**, Machine Learning	10/2020 – 05/2023
BS	**Moscow Institute of Physics and Technology**, Applied Mathematics and Physics	09/2016 – 07/2020

Publications

M. Flicke, G. Angrabeit, M. Iyengar, V. Protsenko, I. Shakun, **J. Cicvaric**, B. Kargi, H. He, L. Schuler, L. Scholz, K. Agnihotri, Y. Cao, A. Geiger. — 2025
Scholar Inbox: Personalized Paper Recommendations for Scientists. | ACL Demo, 2025

- https://aclanthology.org/2025.acl-demo.30/

M. Schneider*, **J. Cicvaric***, A. Sauer, A. Geiger, K. Chitta. — 2024
Generative Dataset Distillation: A New Hope? | Workshop on the Dataset Distillation Challenge, ECCV 2024

- **Won Best Paper Award and Ranked 2nd** at the ECCV 2024 Workshop on the Dataset Distillation

Projects

Generative Dataset Distillation (GenDM) | Python, PyTorch, WandB 2023

- Master's thesis project under Prof. Andreas Geiger, merging dataset distillation and generative modelling.
- Worked with generative models such as StyleGAN2 and StyleGAN-XL.
- Conducted extensive research on existing distillation methods and compared proposed techniques.

Laser-hockey RL | Python, PyTorch, OpenAI Gym, NumPy 2021

- Implemented Deep Deterministic Policy Gradient (DDPG) and Twin Delayed DDPG (TD3).

CarRacing IL & RL | Python, PyTorch, OpenAI Gym, NumPy 2021

- Implemented Deep Q-Network (DQN), Imitation Learning, and a modular geometric controller pipeline for OpenAI Gym's CarRacing environment.